A Necessary and Sufficient Condition for $\pi(I - P + K)$ to be Generalized Drazin Invertible

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Status

This packet gives a likely valid full answer to the open question appearing in the example before Section 3 of Hamdan–Berkani [1]. In that example $P \in L(X)$ is an idempotent, $K \in L(X)$ is compact, and

$$T = I - P + K.$$

The paper gives a sufficient commutant condition for $\pi(T)$ to be generalized Drazin invertible in $\mathcal{C}_0(X) = L(X)/F_0(X)$, then asks more generally for conditions on P and K. The result below shows that the stated sufficient condition is also necessary for this $T = I - P + K$ setup.

Source Question

The source passage occurs in the example immediately after Remark 2.13, around line 847 of the cached source for arXiv:2401.01390. In text form:

> More generally, what are the conditions on P and K so that $\pi(T)$ is generalized Drazin invertible in $\mathcal{C}_0(X)$?

The surrounding example has fixed $T = I - P + K$, where P is idempotent and K is compact.

As P is associated to T in $\mathcal{C}(X)$, then $\Pi(T)+\Pi(P)$ is invertible in $\mathcal{C}(X)$ and so $\pi(T)+\pi(P)$ is invertible in $\mathcal{C}_0(X)$. Hence there exists $S \in L(X)$ such that $\pi(S)(\pi(T)+\pi(P))) = (\pi(T)+\pi(P)))\pi(S) = \pi(I)$. Then $\pi(S)\pi(P) = \pi(P)))\pi(S)$ and

$$S_2T_2 = I_2 + F_1,\ T_2S_2 = I_2 + F_2,$$

where F_1, F_2 are finite rank operators in $L(X_2)$. Therefore T_2 is a Fredholm operator.

3) $\Rightarrow$ 1) Assume that there exists an idempotent $P \in L(X)$ such that $\pi(P) \in comm^2(\pi(T))$, T_1 is a Riesz operator and T_2 is a Fredholm operator. From Theorem 3.9 and Theorem 3.10, it follows that $\pi(T)$ is left generalized invertible and right generalized invertible in $\mathcal{C}_0(X)$. Then from Theorem 3.6, it follows that $\pi(T)$ is generalized invertible in $\mathcal{C}_0(X)$.
As $\pi(P) \in comm^2(\pi(T))$, then $T = T_1 \oplus T_2 + F$, F being a finite rank operator. So T is a finite rank perturbation of a Riesz-Fredholm operator. □

It follows from the properties of generalized Drazin invertibility in a Banach algebra, that if $T \in L(X)$ is a pseudo B-Fredholm operator and if P is an idempotent associated to T in $\mathcal{C}(X)$, then $comm(\Pi(T)) \subset comm(\Pi(P))$. This a direct consequence of the properties of the functional calculus, see [14, Theorem 3.1] for the construction of the idempotent $p = \Pi(P)$.

Remark 3.12. Let $T \in L(X)$ be a pseudo B-Fredholm operator and let P be an idempotent associated to T in $\mathcal{C}(X)$. Then $\pi(T)$ is generalized Drazin invertible in $\mathcal{C}_0(X)$ if and only if $comm(\pi(T) \subset comm(\pi(P))$. Indeed $\pi(P) \in comm^2(\pi(T))$ is equivalent to $comm(\pi(T)) \subset comm(\pi(P))$ and then the remark is consequence of Theorem 3.11.

Example 3.13. Let P be an idempotent element of $L(X)$ and let K be a compact operator. Consider the operator $T = I - P + K$. Then T is a pseudo B-Fredholm operator with P as an associated idempotent in $\mathcal{C}(X)$. From [14, Theorem 4.2], the idempotent $p = \Pi(P)$ is the unique idempotent in $\mathcal{C}(X)$ such that $\Pi(T)p = p\Pi(T)$, $\Pi(T) + p$ is invertible and $\Pi(T)p$ is quasi-nilpotent.

Now if there exists an idempotent Q in $L(X)$ such that $\pi(T)$ is generalized Drazin invertible in $\mathcal{C}_0(X)$ and Q is associated to T in $\mathcal{C}_0(X)$, then Q is associated to T in $\mathcal{C}(X)$. So we must have $\Pi(Q) = \Pi(P)$. In particular if $comm(\pi(T)) \subset comm(\pi(K))$, then $comm(\pi(T)) \subset comm(\pi(P))$. From Remark 3.12 it follows that $\pi(T)$ is generalized Drazin invertible in $\mathcal{C}_0(X)$ and P is associated to T in $\mathcal{C}_0(X)$.

Open Question: More generally, what are the conditions on P and K so that $\pi(T)$ is generalized Drazin invertible in $\mathcal{C}_0(X)$?

4 Index of Fredholm type Operators

In this section, we will define an index for pseudo-B-Fredholm operators. As we will see, this new index extends naturally the usual Fredholm index and satisfies similar properties.

Definition 4.1. *Let $T \in L(X)$ be a pseudo semi-B-Fredholm operator and let P be an idempotent element in $L(X)$ associated to T in $\mathcal{C}(X)$. We define the index **ind**(T) of T as the index of the semi-Fredholm operator $T + P$, that's **ind**(T)= ind$(T + P)$.*

Theorem 4.2. *Let $T \in L(X)$ be a pseudo semi-B-Fredholm operator. Then the index ind(T) of T is well defined and **ind**(T)= $\lim\limits_{\lambda \to 0}$ ind$(T - \lambda I)$.*

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Definitions and Source Input

Let X be a Banach space. Write

$$\pi : L(X) \to \mathcal{C}_0(X) = L(X)/F_0(X)$$

for the quotient map by the ideal of finite-rank operators, and write

$$\Pi : L(X) \to \mathcal{C}(X) = L(X)/K(X)$$

for the Calkin quotient by compact operators. For an element a in an algebra, $\mathrm{comm}(a)$ denotes its commutant.

We use the following result from the source paper.

Lemma 1 (Hamdan–Berkani, Remark 2.13)**.** *Let $S \in L(X)$ be a pseudo B-Fredholm operator, and let P be an idempotent associated to S in the Calkin algebra $\mathcal{C}(X)$. Then $\pi(S)$ is generalized Drazin invertible in $\mathcal{C}_0(X)$ if and only if*

$$\mathrm{comm}(\pi(S)) \subseteq \mathrm{comm}(\pi(P)).$$

In the source example with $T = I - P + K$, K compact, the paper observes that T is pseudo B-Fredholm and that P is associated to T in $\mathcal{C}(X)$. Thus Lemma 1 applies to this T.

Proof Intuition

Only one algebraic identity is missing from the source's sufficient-condition argument. In $\mathcal{C}_0(X)$, set

$$t = \pi(T), \qquad p = \pi(P), \qquad k = \pi(K).$$

Since $T = I - P + K$, we have $t = 1 - p + k$, equivalently $p = 1 + k - t$. Therefore, for every x commuting with t,

$$xp - px = x(1 + k - t) - (1 + k - t)x = xk - kx.$$

So, on the commutant of t, commuting with p is exactly the same as commuting with k.

Main Theorem

Theorem 1. *Let X be a Banach space, $P \in L(X)$ an idempotent, $K \in L(X)$ a compact operator, and $T = I - P + K$. Put $t = \pi(T)$ and $k = \pi(K)$. Then*

$$t \text{ is generalized Drazin invertible in } \mathcal{C}_0(X) \quad \Longleftrightarrow \quad \mathrm{comm}(t) \subseteq \mathrm{comm}(k).$$

Equivalently,

$$\begin{gathered}\pi(I - P + K) \text{ is generalized Drazin invertible in } \mathcal{C}_0(X) \\ \Longleftrightarrow \quad \mathrm{comm}(\pi(I - P + K)) \subseteq \mathrm{comm}(\pi(P)).\end{gathered}$$

Proof. Let $p = \pi(P)$. The source example shows that T is pseudo B-Fredholm and that P is associated to T in $\mathcal{C}(X)$. By Lemma 1,

$$t \text{ is generalized Drazin invertible in } \mathcal{C}_0(X) \quad \Longleftrightarrow \quad \mathrm{comm}(t) \subseteq \mathrm{comm}(p).$$

It remains only to compare the two commutant inclusions. Since

$$t = \pi(I - P + K) = 1 - p + k,$$

we have $p = 1 + k - t$. If $x \in \mathrm{comm}(t)$, then

$$xp - px = x(1 + k - t) - (1 + k - t)x = xk - kx.$$

Hence, for every $x \in \mathrm{comm}(t)$, $x \in \mathrm{comm}(p)$ if and only if $x \in \mathrm{comm}(k)$. Therefore

$$\mathrm{comm}(t) \subseteq \mathrm{comm}(p) \quad \Longleftrightarrow \quad \mathrm{comm}(t) \subseteq \mathrm{comm}(k).$$

Combining this equivalence with Lemma 1 proves the theorem. □

Consequences

The sufficient condition stated in the source,

$$\mathrm{comm}(\pi(T)) \subseteq \mathrm{comm}(\pi(K)),$$

is not merely sufficient for $T = I-P+K$; it is necessary and sufficient. In particular, the obstruction can be stated entirely in terms of P and K: every operator that commutes with $I-P+K$ modulo finite rank must commute with K modulo finite rank, equivalently with P modulo finite rank.

Verification Notes

The proof has no computational component. The only external mathematical input is the source paper's Remark 2.13, together with the source example's observation that $T = I-P+K$ is pseudo B-Fredholm with P associated to T in the Calkin algebra. The added argument is the identity $[x, \pi(P)] = [x, \pi(K)]$ for all $x \in \mathrm{comm}(\pi(T))$.

Limitations

This theorem answers the open question in the natural scope of the displayed source example $T = I - P + K$. It does not attempt to classify arbitrary compact perturbations of unrelated pseudo B-Fredholm decompositions beyond that normal form.

Novelty Check

The local run indexes were searched for arXiv:2401.01390, the source title, "conditions on P and K", "$I - P + K$", "generalized Drazin invertible", and the commutant-inclusion phrasing. No duplicate packet or attempt was found. A bounded web search on June 29, 2026 for the exact open question language, the source title, and the phrase "$T = I - P + K$" with "generalized Drazin invertible" found no later explicit answer.

Human Review Recommendation

Review the interpretation of the open question as referring to the immediately preceding $T = I-P+K$ setup. Under that reading, the proof is complete and the necessary direction is the only point not written in the source example.

References

[1] A. Hamdan and M. Berkani, *Fredholm-type Operators and Index*, arXiv:2401.01390, 2024.

[2] J. J. Koliha, *A generalized Drazin inverse*, Glasgow Mathematical Journal 38 (1996), 367–381.